

[54] **PROCESS FOR THE PRODUCTION OF CLAVULANIC ACID**

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[58] Field of Search **195/80 R**

[56] **References Cited**

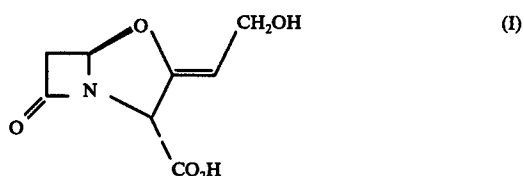
U.S. PATENT DOCUMENTS

3,770,590	11/1973	Inamine et al.	195/80 R
3,801,464	4/1974	Gorman et al.	195/80 R
3,862,008	1/1975	Hamill et al.	195/80 R

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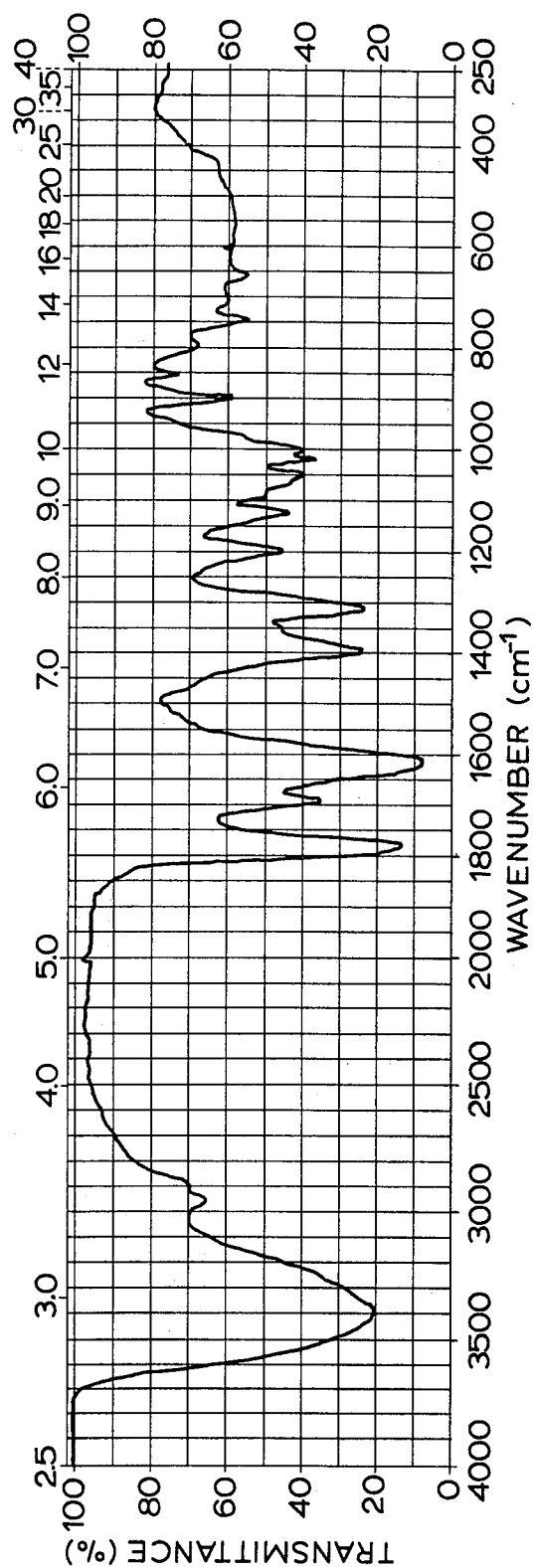
[57] **ABSTRACT**

A new antibacterially active agent has been isolated from *Streptomyces clavuligerus*. This new compound which is designated clavulanic acid has the formula (I):



In addition to being a broad spectrum antibiotic of medium potency, clavulanic acid and its salts and esters have the ability to enhance the effectiveness of β -lactam antibiotics against many β -lactamase producing bacteria.

19 Claims, 1 Drawing Figure



PROCESS FOR THE PRODUCTION OF CLAVULANIC ACID

CROSS REFERENCE

This is a division of application Ser. No. 569,007, filed Apr. 17, 1975.

BACKGROUND TO THE INVENTION

a. *Streptomyces clavuligerus* has been described in detail by Higgins et al., *Int.J.Systematic Bacteriology*, 21, 326 (1971). This streptomycete was of interest because it produced certain β -lactam antibiotics such as penicillin N, 7-(5-amino-5-carboxyvaleramido)-3-carbamoyloxymethyl-3-cephem-4-carboxylic acid and 7-(5-amino-5-carboxyvaleramido)-3-carbamoyloxymethyl-7-methoxy-3-cephem-4-carboxylic acid. The streptomycete has been deposited in the Agricultural Research Service Collection as NRRL 3585 and in the American Type Cultural Collection as ATCC 27064. *Streptomyces clavuligerus* has also been referred to in U.S. Pat. No. 3,770,590 and also by Nagarajan et al., *J.Amer.Chem.Soc.*, 93, 2308 (1971), Brannon et al., *Antimicrob. Agents Chemother.*, 1, 237 (1972) and *Antimicrob. Agents Chemother.*, 1, 247 (1972) and Higgins et al., *J. Antibiot.*, 27, 298 (1974).

b. β -lactamases are enzymes which open the β -lactam ring of penicillins and cephalosporins to give products which are devoid of antibacterial activity. These enzymes are produced by many bacteria, notably species or strains of *Escherichia*, *Klebsiella*, *Proteus*, *Pseudomonas*, *Enterobacter* and *Staphylococcus* and are in many instances the explanation for the resistance of certain strains of such organisms to some penicillins and cephalosporins. The importance of β -lactamase production may be understood when it is realised that a high proportion of clinically isolated organisms produce β -lactamases (see, for example, M. Wilson and I. A. Freeman, *Bacteriological Proceedings*, 80 (1969) where in a paper entitled 'Penicillin Inactivation by Gram-negative Bacilli' they showed that 84% of the gram-negative organisms isolated in an American hospital produced β -lactamase). In many cases, some penicillins or cephalosporins are ineffective in treating diseases ascribed to non β -lactamase-producing organisms because of the common occurrence of co-infection by a β -lactamase producer (see, for example, R. May et al.; *Brit. J.Dis.Chest.*, 66, 185 (1972)). Combination of a β -lactamase inhibiting substance with a penicillin or cephalosporin might be expected to protect the latter from degradation by bacterial β -lactamase and thereby enhance their antibacterial activity against many infective organisms. This process of enhancement of the antibacterial activity is called synergism when the antibacterial activity of the combination is well in excess of the simple addition of the activities of the two separate substances. The β -lactamase inhibiting component of the mixture is referred to as a synergist and such substances are valuable for increasing the antibacterial activity of penicillins and cephalosporins against resistant organisms. It is one of the objects of this invention to provide such synergists.

c. Examples of the use of certain β -lactamase resistant semi-synthetic penicillins and cephalosporins as β -lactamase inhibitors and synergists for penicillins and cephalosporins have already been described in the literature, see for example, Sutherland et al., *Nature*, 201, 868 (1964); Sabath et al., *Nature*, 204, 1066 (1964); O'Callag-

han et al., *Antimicrob. Agents and Chemotherapy*, 1968, 67 (1969). However, none of these known agents have a dramatic effect on the spectrum of the other antibiotic present in the mixture.

d. Certain actinomycete cultures have been described as producing β -lactamase inhibiting substances which act synergistically with penicillins or cephalosporins, for example, those cultures disclosed in British Patent No. 1,363,075 and those described by Hata et al., *J. Antibiotics*, 25, 473 (1972) and Umezawa et al., *J. Antibiotics*, 26, 51 (1973). None of these β -lactamase inhibitors of actinomycetal origin have yet been found to be of use in the clinic. Particularly noteworthy features which distinguish clavulanic acid from other β -lactamase inhibitors of actinomycetal origin are its extractability into organic solvents from culture filtrate at pH2, its high stability in human blood and its broad spectrum of anti-bacterial and β -lactamase inhibiting activity, its low molecular weight and its high R_f values on paper chromatography using a variety of solvent systems.

DESCRIPTION OF THE INVENTION

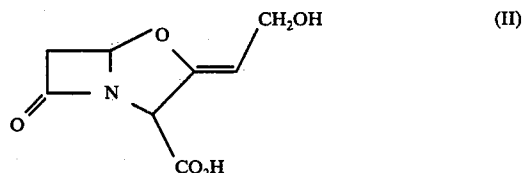
We have discovered that the aerobic cultivation of *Streptomyces clavuligerus* in conventional nutrient media at about 25°-30° C under roughly neutral conditions produces a β -lactamase inhibitory substance which also possesses antibacterial activity. We have designated this new material 'clavulanic acid'.

Clavulanic acid has the following properties:

- It is a carboxylic acid.
- It forms a sodium salt which has a characteristic infra-red spectrum substantially as shown in the drawing.
- It is able to inhibit the growth of strains of *Staphylococcus aureus*.
- It is able to synergise the antibacterial effect of ampicillin against β -lactamase producing strains of *Escherichia coli*, *Klebsiella aerogenes* and *Staphylococcus aureus*.
- It is able to synergise the antibacterial effect of cephaloridine against the β -lactamase producing strains of *Proteus mirabilis* and *Staphylococcus aureus*.
- It forms a methyl ester which has a molecular weight (by mass spectroscopy) of 213.0635 which corresponds to the formula $C_9H_{11}NO_5$.

Thus clavulanic acid may be regarded as a monobasic carboxylic acid of the formula $C_9H_9NO_5$ which in the form of its sodium salt has a characteristic infra-red absorption spectrum substantially as shown in the drawing.

The compound produced by *Streptomyces clavuligerus* which has the above properties has the formula (II):



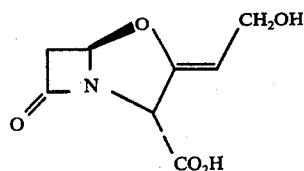
Thus clavulanic acid may be named 3-(β -hydroxyethylidene)-7-oxo-4-oxa-1-azabicyclo[3,2,0]heptane-2-carboxylic acid.

The stereochemistry at C_5 and C_2 of the clavulanic acid is the same as that found in naturally occurring

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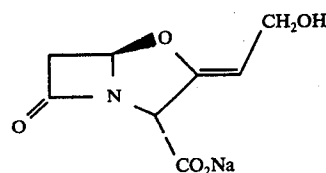
penicillins and cephalosporins so that clavulanic acid may be represented by the structural formula (I):



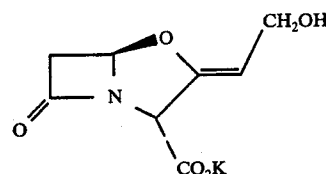
(I) 5

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(III)



(IV)

Thus a fuller chemical name for clavulanic acid is Z-(2R,5R)-3-(beta-hydroxyethylidene)-7-oxo-4-oxa-1-azabicyclo[3,2,0]heptane-2-carboxylic acid.

The great usefulness of clavulanic acid may be readily appreciated when it is realised that certain strains of *Klebsiella aerogens* A, the growth of which is not inhibited by the presence of 125 µg/ml. of ampicillin, amoxycillin, carbenicillin or benzyl penicillin or by the presence of 10 µg/ml. of clavulanic acid, are inhibited by the presence of less than 12.5 µg/ml. of the previously mentioned penicillins when 5 µg/ml. of clavulanic acid is also present. Similar results have been observed for combinations containing various esters of clavulanic acid. For example, strains of *Klebsiella aerogens* A, the growth of which is not inhibited by 125 µg/ml. of ampicillin, or by 10 µg/ml of clavulanic acid methyl ester are inhibited by less than 12.5 µg/ml. of ampicillin in the presence of 5 µg/ml. of the clavulanic acid methyl ester. It has also been found that strains of *Staphylococcus aureus* Russell, the growth of which is not inhibited by the presence of 100 µg/ml. of ampicillin or by 5 µg/ml of clavulanic acid, are inhibited by the presence of less than 10 µg/ml. of ampicillin in the presence of 1 µg/ml. of clavulanic acid. In tests on female mice, it has been found that blood and tissue levels of clavulanic acid considerably in excess of 5 µg/ml. can readily be achieved by subcutaneous administration of 100 mg/kg of the sodium salt of clavulanic acid and that useful levels of clavulanic acid can be obtained after oral administration of 100 mg/kg of the sodium salt of clavulanic acid.

Accordingly, the present invention provides clavulanic acid as hereinbefore described and its salts and esters.

Most suitably the salts of clavulanic acid will be pharmaceutically acceptable salts such as the sodium, potassium, calcium, magnesium, aluminium, ammonium and substituted ammonium salts such as the trimethylammonium, benzathine, procaine and like salts conventionally formed with penicillins or cephalosporins. Non-pharmaceutically acceptable salts of clavulanic acid are also included within the scope of this invention as they as well as the pharmaceutically acceptable salts are useful intermediates in the preparation of esters of clavulanic acid, for example, the sodium lithium or silver salts of clavulanic acid may be reacted with benzyl bromide to form the useful benzyl ester of clavulanic acid.

Salts of clavulanic acid tend to be more stable than the parent acid per se and thus form a favoured aspect of this invention. Particularly suitable salts of clavulanic acid include the sodium and potassium salts which have the formula (III) and (IV) respectively:

Crystalline forms of such salts may contain water of hydration.

Suitable esters of clavulanic acid include those notionally derived from alcohols such as methanol, ethanol, propanol, butanol, 2,2,2-trichloroethanol, 2,2,2-trifluoroethanol, benzyl alcohol, p-nitrobenzyl alcohol, phenol, acetoxymethanol, pivaloyloxymethanol, 2-dimethylaminoethanol and other conventional alcohols. Various esters of clavulanic acid are useful intermediates in certain processes for the purification of clavulanic acid. Many clavulanic acid esters are useful synergistic compounds. The activity of such esters might be due to hydrolysis of the ester to the parent acid.

When used herein the term ester includes esters notionally derived from an alcohol or thiol of the formula ROH or RSH where R is an organic residue. Suitable groups R include alkyl, alkenyl, alkynyl, aryl, arylalkyl or other similar groups any of which may be substituted if desired. In order not to increase the molecular weight to an unreasonable extent, groups R do not normally include more than 16 carbon atoms, more suitably, not more than 12 carbon atoms and most suitably, not more than 8 carbon atoms.

Preferably, the group R is notionally derived from an alcohol ROH or (less favourably) a thiol RSH which is pharmaceutically acceptable.

Suitable substituents which may be included in the group R include halogen atoms and lower alkoxyl, hydroxyl, lower acyloxy, lower alkylamino, lower dialkylamino and like groups. The term "lower" means that the group contains up to 6 carbon atoms, and preferably up to 4 carbon atoms. Thus, for example, R may be a methyl, ethyl, n-propyl, iso-propyl, straight or branched butyl, pentyl, heptyl, octyl, nonyl, decyl, undecyl, dodecyl, vinyl, allyl, butenyl, cyclopropyl, cyclobutyl, cyclopentyl, cyclohexyl, cycloheptyl, cyclohexenyl, cyclohexadienyl, methylcyclopentyl, methylcyclohexyl, benzyl, benzhydryl, phenylethyl, naphthylmethyl, phenyl, naphthyl, propynyl, tolyl, 2-chloroethyl, 2,2,2-trichloroethyl, 2,2,2-trifluoroethyl, acetylmethyl, benzoylmethyl, 2-methoxyethyl, 2-dimethylaminoethyl, 2-diethylaminoethyl, 2-piperidinoethyl, 2-morpholinoethyl, 3-dimethylaminopropyl, p-chlorobenzyl, p-methoxybenzyl, p-nitrobenzyl, p-bromobenzyl, m-chlorobenzyl, 6-methoxynaphthyl-2-methyl, p-chlorophenyl, p-methoxyphenyl or any like group as well as those groups which are known from the penicillin or cephalosporin arts to produce esters known to be readily hydrolysed in vivo to the parent antibiotic.

Readily hydrolysable esters include, but are not limited to, those of the formulae (V) and (VI):